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Department of Computing

Software Systems

20MSSL10 - Business Intelligence Laboratory

Abstract

Title : ProcureIQ: Intelligent Procurement Analytics and Investigation System

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Problem Statement

Organizations generate large volumes of procurement data related to suppliers, products, contracts, prices, purchase values, and dates. However, procurement teams often lack an integrated system to convert this data into meaningful and actionable insights.

The proposed system analyzes procurement spending, supplier performance, price trends, and purchasing patterns. It also uses predictive analytics to forecast demand and identify potential savings opportunities. An AI-powered Procurement Investigation Agent further reduces the manual effort required to investigate abnormal prices, supplier issues, and procurement trends.

Dataset to be Used

Primary Dataset: GTI Global Public Procurement Dataset (GPPD)

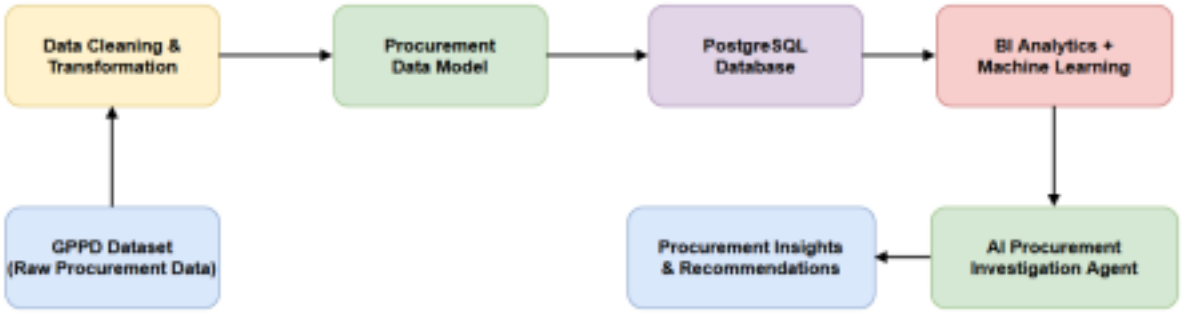
The GTI Global Public Procurement Dataset (GPPD) will be used as the primary real-world procurement dataset. It contains procurement contract information from **multiple countries, including buyers, suppliers, products or categories, contract values, dates, procurement procedures, and geographic information.**

The dataset will be used to analyze **supplier activity, procurement history, product purchasing patterns, contract values, price trends, supplier spending, procurement volume, and supplier or product comparisons.**

Dataset Workflow

The **GPPD** dataset first undergoes data cleaning and transformation to remove inconsistencies and prepare the data for analysis. The processed information is organized into a procurement data model and stored in a **PostgreSQL** database.

The database then supports BI dashboards and machine-learning models for spend analysis, supplier analytics, price anomaly detection, and demand forecasting. The AI Procurement Investigation Agent accesses only the relevant records and analytics results required for a particular investigation rather than receiving the complete raw dataset.



Important BI Features

1. Procurement Spend Analysis

Analyzes total spending across suppliers, products, categories, and time periods to identify major expenditure areas.

2. Purchase Volume Analysis

Tracks procurement volume across suppliers, products, and time periods to identify purchasing patterns and changes.

3. Supplier Analytics

Evaluates suppliers based on spending, contract activity, pricing, performance, and historical trends.

4. Price Trend Analysis

Monitors product prices over time and compares pricing patterns across suppliers and categories.

5. Price Anomaly Detection

Uses statistical and machine-learning techniques to identify unusually high or low procurement prices.

6. Demand Forecasting

Uses historical procurement patterns to estimate future purchasing requirements and expected demand.

7. Procurement KPI Dashboard

Provides a consolidated view of key procurement indicators such as spend, volume, suppliers, contracts, prices, anomalies, and forecasted demand.

Agent Functions

The AI Procurement Investigation Agent uses relevant procurement records and analytics results to investigate issues and provide evidence-based insights.

1. Price Anomaly Investigation

Investigates abnormal prices using historical prices, supplier comparisons, procurement volume, demand, and performance indicators.

2. Supplier Performance Investigation

Analyzes supplier history, pricing, delivery, quality, and contract information to identify performance changes.

3. Alternative Supplier Identification

Identifies and compares suppliers offering the same or comparable products based on relevant procurement indicators.

4. Potential Savings Analysis

Estimates potential savings by comparing current purchase costs with relevant alternative supplier prices.

5. Procurement Trend Investigation

Investigates unusual changes in procurement spending, purchasing volume, supplier

activity, prices, or demand.

6. Procurement Question Answering

Answers procurement-related questions using structured procurement data, analytics results, and supplier or product information.